Q1: 24 Feb (Shift 1) - Single Correct

Each side of a box made of metal sheet in cubic shape is 'a' at room temperature 'T', the coefficient of linear expansion of the metal sheet is ' α '. The metal sheet is heated uniformly, by a small temperature $\Delta \mathrm{T}$, so that its new temperature is $\mathrm{T}+\Delta \mathrm{T}$. Calculate the increase in the volume of the metal box-

(1) $\frac{4}{3} \pi \mathrm{a}^{3} \alpha \Delta \mathrm{T}$

(2) $4 \pi \mathrm{a}^{3} \alpha \Delta \mathrm{T}$

(3) $3 a^{3} \alpha \Delta T$

(4) $4 \mathrm{a}^{3} \alpha \Delta \mathrm{T}$

Q2: 25 Feb (Shift 1) - Single Correct

A proton, a deuteron and an α particle are moving with same momentum in a uniform magnetic field. The ratio of magnetic forces action on them is ____ and their speed is ____ in the ratio.

(1) 2: 1: 1 and 4: 2: 1

(2) 1: 2: 4 and 2: 1: 1

(3) 1: 2: 4 and 1: 1: 2

(4) 4: 2: 1 and 2: 1: 1

Answer Key

Q1 (3) **Q2** (1)

Q1 (3)

volume expansion $\gamma = 3\alpha$

$\frac{\Delta \mathrm{V}}{\mathrm{V}} = \gamma \Delta \mathrm{T}$

$\Delta \mathrm{V} = \mathrm{V} \cdot \gamma \Delta \mathrm{T}$

$\Delta \mathrm{V} = \mathrm{a}^3 .3\alpha \Delta \mathrm{T}$

Q2 (1)

As $v = \frac{p}{m} \& F = qvB$

$\therefore F = \frac{qp}{m} B$

$F_1 = \frac{qpB}{m}, v_1 = \frac{p}{m}$

$F_2 = \frac{qpB}{2m}, v_2 = \frac{p}{2m}$

$F_3 = \frac{2qpB}{4m}, v_3 = \frac{p}{4m}$

$F_1 : F_2 : F_3 \& V_1 : V_2 : V_3$

$1 : \frac{1}{2} : \frac{1}{2} \quad \& 1 : \frac{1}{2} : \frac{1}{4}$

2 : 1 : 1 \& 4: 2: 1